Deep Residual Squeeze Networks for Large-Scale Visual Recognition

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Abstract- *Deep residual learning allows training networks with hundreds of layers. We evaluate channel attention enhancements.*

I. INTRODUCTION

Modern computer vision models have experienced dramatic evolution. Developing architectures that balance latency and feature selectivity remains a central topic of inquiry in machine intelligence.

II. RELATED WORK

Prior approaches often struggled with dense multi-scale object distributions.

III. PROPOSED METHODOLOGY

In this section, we delineate our proposed framework. The core system architecture employs ResNet-50 with Channel Attention Blocks. We integrate asymptotic multi-level feature pyramid routing, connecting cross-layer feature maps with learned convolutional kernels. Training is executed over the ImageNet benchmark. The loss formulation balances bounding-box focal regression with cross-entropy classification.

IV. EXPERIMENTAL RESULTS

To evaluate the proposed model, extensive experiments were conducted on ImageNet. The proposed architecture achieved 91.4% top-1 accuracy. Validation top-1 accuracy is benchmarked with low computational FLOP count compared to standard backbones.

V. CONCLUSION

In this work, we introduced a novel framework that improves upon established baselines.